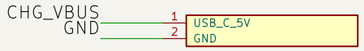
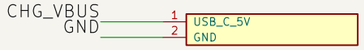


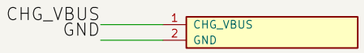

J1
4x 21700 pack (4S)



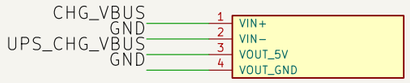

J2
Foldable solar panel



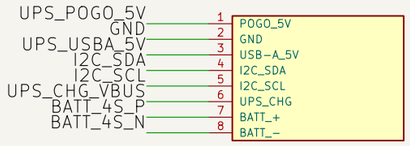

J3
Hand crank generator



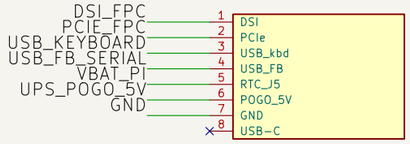

J4
Sealed USB-C port



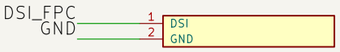

J5
MPPT / charger



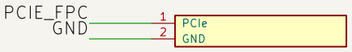

J6
Waveshare UPS HAT (E)




J7
Raspberry Pi 5



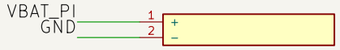

J8
DFR0550-V2 5 inch DSI



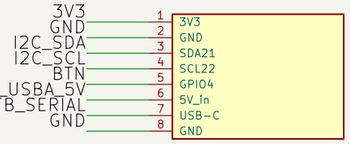

J9
Seeed M.2 BASE + 1TB



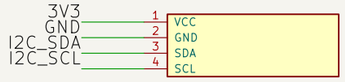

J10
Phistone 2.4G dongle



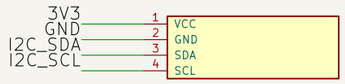

J11
Panasonic ML-2020



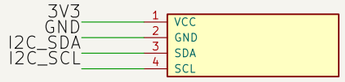

J12
FireBeetle 2 ESP32-UE




J13
1.3 inch OLED 0x3C



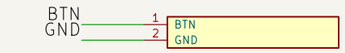

J14
BME688 0x76/0x77




J15
DS3231 RTC 0x68




J16
Trackball 0x0A (r2 only)




J17
PBS-33B status button

Same net label = connected. This is a net-label schematic, not a routed harness.

I2C: FireBeetle is the only master (3.3 V). Fit one set of 4.7 kOhm pull-ups only.

CHG_VBUS: solar and crank share the one sealed port. BATT_4S: 4x21700 series, fuse positive.

USB_FB_SERIAL: FireBeetle USB-C to Pi USB-A, data-only preferred. Trackball is r2-only; r3 uses none.

R3 keyboard: the case holds only the Phistone 2.4G dongle (J10) in a Pi USB-A port. Bluetooth option: no dongle, no wire.

**Ktemata**

Sheet: /
File: ktemata_r3_wiring.kicad_sch

***Title: Ktemata r3 — wiring / connections***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: 2** |
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |